MANDALA: A Modular E(3)-Equivariant Framework for Learning Sparse Electronic-Structure Matrices and Downstream Observables

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Abstract

We present Mandala, a modular software framework for learning block-sparse electronic-structure matrices with E(3)-equivariant graph neural networks. The framework is designed around a unified representation of atom-resolved Hamiltonian, overlap, and density matrices, together with reusable abstractions for basis conversion, sparse block handling, irrep mapping, graph construction, model definition, and training. This design enables Mandala to support multiple electronic-structure backends, heterogeneous chemical compositions, and a wide range of neural architecture variants within one workflow.

A distinguishing feature of Mandala is that it does not stop at matrix reconstruction. Predicted matrices can be used directly to evaluate observable-level quantities such as total energies and electron counts, and the predicted Hamiltonian also preserves access to band-structure analysis through standard electronic-structure post-processing. The framework supports training with observable-based losses. Through automatic differentiation of observables with respect to atomic coordinates, the same formulation extends naturally to force supervision. This allows the software to bridge the gap between electronic-structure learning and observable-guided atomistic modeling while retaining a representation tied to quantum-mechanical operators rather than

only scalar targets.

Mandala combines a modular software architecture with an operator-centered learning formulation for sparse electronic-structure prediction. We emphasize the sparse matrix data model, the observable-aware loss design, and the broad hyperparameter and architecture search space available in the framework. Numerical results for evaluated systems [TODO: list evaluated systems] are provided in Section [TODO: numerical results section].

Keywords: Electronic structure, Density functional theory, Equivariant graph neural networks, Sparse matrix learning, Energy and force prediction, Scientific machine learning

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PROGRAM SUMMARY

Program Title: Mandala

CPC Library link to program files: [TODO: to be added by the Technical Editor]

Developer's repository link: `https://github.com/Mandala-org/Mandala`

Licensing provisions: BSD-3-Clause

Programming language: Python

Nature of problem: Electronic-structure calculations based on Kohn-Sham density functional theory remain a central tool in computational materials science and chemistry, but their computational cost limits accessible system sizes, trajectory lengths, and hyperparameter studies. Machine-learned interatomic potentials alleviate this cost for energies and forces, yet they generally do not expose the electronic operators needed for richer observables or electronic post-processing. A complementary route is to learn sparse representations of electronic-structure matrices themselves. Such a program must support multiple atom types, changing orbital bases, periodic boundary conditions, equivariant geometric processing, and physically meaningful observable targets such as total energy and atomic forces.

Solution method: Mandala represents Hamiltonian, overlap, and density matrices as atom-pair-resolved sparse block tensors and maps these blocks to irreducible E(3) representations compatible with equivariant neural networks. The package combines parser modules for electronic-structure data, basis-conversion utilities, a shared irrep mapper, graph construction routines, and configurable E(3)-equivariant message-passing models. Multi-head prediction allows simultaneous learning of several matrices, while observable-level quantities are evaluated from the predicted operators using sparse traces. Observable losses can be added during training, and force supervision is obtained through automatic differentiation of predicted energies with respect to atomic coordinates. The framework is organized to support extensibility, architecture search, and reproducible training workflows.

Additional comments including restrictions and unusual features: The framework is optimized for sparse matrix learning in atomistic systems with localized orbital bases and periodic or non-periodic geometries. A distinctive feature of Mandala is the combination of operator-level learning with optional observable guidance derived from those operators, which enables a single software stack to serve both electronic-structure emulation and observable-guided atomistic applications. The package is designed to accommodate multiple data backends and a wide search space of equivariant architectures and training objectives.

1. Introduction

Kohn-Sham density functional theory (DFT) is one of the standard workhorses of electronic-structure simulation because it offers a practical compromise between physical fidelity and computational cost across chemistry, condensed-matter physics, and materials science [1, 2]. Even so, the repeated solution of the Kohn-Sham equations remains a major bottleneck in workflows that require large supercells, long molecular-dynamics trajectories, broad compositional screening, or repeated re-evaluation under changing thermodynamic conditions. This cost has motivated sustained interest in machine-learned surrogates for quantum-mechanical simulation, including direct learning of densities, wave-function-related quantities, and reduced electronic representations [3–6].

Many successful atomistic machine-learning approaches bypass the explicit electronic structure and instead learn scalar or vector observables such as energies and forces. This strategy is highly effective for interatomic potentials, and recent work has shown that equivariant neural architectures are especially well suited for energy and force prediction because they can encode rotational structure directly in the model rather than relying entirely on handcrafted invariants or data augmentation [7, 8]. However, learning only energies and forces also limits what the surrogate can expose: electronic observables, basis-dependent quantities, operator-level post-processing, and band-structure information derived from the Hamiltonian are no longer directly available. For applications that remain fundamentally electronic-structure-centered, it is attractive to learn the operators themselves.

Mandala is designed around this operator-learning viewpoint. The framework targets block-sparse matrix quantities arising in localized-orbital electronic-structure calculations, in particular the Hamiltonian, overlap, and density matrices. Instead of treating these objects as dense global arrays, Mandala represents them as atom-pair-resolved sparse blocks and combines this representation with E(3)-equivariant graph neural networks. This yields a learning problem that respects both the locality of the underlying electronic structure and the geometric transformation laws imposed by three-dimensional space.

The software is motivated by three specific design goals. First, it should be modular at the level of both scientific abstractions and code organization. Data ingestion, basis conversion, graph construction, irrep mapping, model definition, and training logic should be independently replaceable while still composing into a single workflow. Second, the framework should support supervision beyond raw matrix reconstruction. When Hamiltonian, overlap, and density matrices are predicted jointly, physically relevant observables such as total energies and electron counts can be computed directly from the predicted operators, and atomic forces can be obtained through differentiation with respect to atomic coordinates. Third, the framework should make architecture exploration a first-class capability rather than an afterthought. Choices such as irreducible feature content, message-passing depth, tensor-product design, head structure, residual connections, normalization, and observable losses must all be accessible through a coherent configuration and sweep interface.

These goals distinguish Mandala from earlier machine-learning workflows in two ways. Relative to purely observable-driven interatomic models, Mandala preserves an operator-level representation that remains useful for electronic-structure analysis. Relative to monolithic research scripts for one specific dataset or one fixed neural architecture, it provides a reusable framework in which data formats, physical targets, and model variants can be exchanged without rewriting the entire code path. In this sense, Mandala is intended as a unified software environment for sparse operator learning, observable guidance, and architecture exploration within a single framework.

The following sections describe the sparse matrix data model, the E(3)-equivariant learning formulation, the observable-aware loss construction, the force-training pathway, and the software abstractions that make these components composable. Numerical benchmarks on [TODO: list benchmark systems] and application-specific results for [TODO: list target applications] are reported in Section [TODO: numerical results section].

2. Theoretical Background

2.1. Electronic structure problem and density functional theory

Under the Born-Oppenheimer approximation, the electronic structure problem is posed for fixed nuclear coordinates $\{\boldsymbol{R}_I\}$, while the electronic ground state is obtained from the many-electron Hamiltonian [9]. In practice, the Kohn-Sham formulation of DFT replaces the interacting many-body problem by a set of one-electron equations whose self-consistent solution yields the ground-state density and total energy [1, 2]. For a set of Kohn-Sham orbitals $\{\psi_n\}$ and occupations $\{f_n\}$, the density is

$$\rho(\boldsymbol{r}) = \sum_n f_n |\psi_n(\boldsymbol{r})|^2, \tag{1}$$

and the orbitals satisfy

$$\left[-\frac{1}{2}\nabla^2 + v_{\text{ext}}(\boldsymbol{r}) + v_{\text{H}}[\rho](\boldsymbol{r}) + v_{\text{xc}}[\rho](\boldsymbol{r})\right] \psi_n(\boldsymbol{r}) = \varepsilon_n \psi_n(\boldsymbol{r}). \tag{2}$$

In localized atomic-orbital bases, these equations are naturally represented in terms of matrices. The Hamiltonian matrix H, overlap matrix S, and density matrix D contain the information required for many observables, including the electronic energy and electron count. In particular, traces over operator products provide compact expressions such as

$$E \propto \mathrm{Tr}(DH), \qquad N_e = \mathrm{Tr}(DS), \tag{3}$$

up to the conventions used by the underlying electronic-structure code and the decomposition of total-energy contributions. This operator viewpoint is important for Mandala because it makes matrix prediction directly actionable: once H, S, and D are learned, scalar observables can be recovered without introducing an additional surrogate model.

Localized orbital bases also induce a sparsity structure that is favorable for machine learning. Because electronic matter is spatially local or "nearsighted" in many systems [10], matrix elements between distant local orbitals tend to become small, and atom-pair blocks outside a finite cutoff can often be neglected. Instead of learning a dense matrix of dimension equal to the full basis size, one may therefore learn a structured set of local blocks indexed by atom pairs and lattice shifts. This provides a route to scalable electronic-structure surrogates that is compatible with graph-based geometric learning.

The resulting learning problem is still nontrivial. Matrix blocks are basis dependent, transform nontrivially under rotations, and couple orbitals with different angular character. Moreover, periodic electronic-structure data generated by different simulation packages may follow different basis-ordering conventions. A practical framework must therefore combine physical locality with explicit representation-theoretic bookkeeping. This is precisely the role of the sparse block and irrep abstractions used in Mandala.

2.2. E(3)-equivariant graph neural networks for atomistic learning

Atomistic systems are naturally described by three-dimensional geometry together with discrete chemical identities. Graph neural networks use this structure by treating atoms as nodes and pairwise interactions within a cutoff as edges. In standard message passing, node and edge features are iteratively updated by aggregating information over neighbors, allowing the model to construct increasingly nonlocal descriptors from local geometric inputs.

For physical systems in three dimensions, however, geometric consistency is crucial. Scalar quantities should remain invariant under rigid rotations, while vector- and tensor-like quantities should transform covariantly. E(3)-equivariant neural networks enforce these transformation laws at the architectural level by expressing features in terms of irreducible representations of the Euclidean group and by combining them through equivariant operations such as tensor products, spherical harmonics, and parity-aware nonlinearities [7, 8]. In practice, this means that the network learns with features labeled by angular momentum ℓ and parity, rather than by unrestricted channels with no geometric meaning.

This design is particularly well matched to electronic-structure learning in localized orbital bases. Atomic orbitals already carry angular-momentum information, and matrix elements between orbital blocks inherit symmetry constraints from the underlying basis. An E(3)-equivariant network can therefore operate in a feature space that mirrors the structure of the physical objects it predicts. Rather than reducing all geometry to invariant distances or handcrafted descriptors, the network can keep directional information explicitly available throughout message passing and readout.

Equivariant models have already become central in modern atomistic machine learning because they provide a favorable balance between data efficiency, physical inductive bias, and predictive accuracy [7]. For Mandala, the relevant lesson is not only that equivariant architectures improve force-field learning, but also that they provide a principled route to predicting operator-valued targets whose components belong to different rotational channels. This is why the Mandala framework is built around E(3)-equivariant message passing from the outset.

2.3. From equivariant local features to sparse matrix and observable prediction

Mandala bridges sparse electronic-structure matrices and equivariant graph learning by decomposing the problem into three stages. First, electronic-structure data are represented as block-sparse matrices indexed by atom pairs and, for periodic systems, lattice-image shifts. Second, each block is mapped to an irrep vector whose components transform according to the angular structure implied by the associated orbital pair. Third, an equivariant graph neural network predicts these irrep vectors from atomic geometry and species information, after which the vectors are mapped back to matrix blocks.

This formulation has two important consequences. The first is modularity of representation: basis handling, sparse indexing, equivariant learning, and observable evaluation can be implemented as separate software layers. The second is physical extensibility of supervision. Once the network predicts a consistent set of matrices, observables need not be learned independently; they can be evaluated from the predicted operators. In the Mandala workflow, total-energy-like scalars are evaluated from sparse traces involving the predicted Hamiltonian and density matrices, electron counts from density-overlap traces, and forces from

$$\boldsymbol{F}_I = -\frac{\partial E}{\partial \boldsymbol{R}_I}, \tag{4}$$

obtained through automatic differentiation of the predicted energy with respect to atomic coordinates.

This operator-first viewpoint differs from the more common direct-regression route in which separate network heads are trained independently for energies and forces. The advantage of the Mandala strategy is that the predicted quantities remain internally connected: energies, electron counts, and forces are derived from the same learned electronic representation. This gives the user a single framework that can be trained for matrix fidelity, observable fidelity, or both, depending on the scientific objective.

3. Numerical Implementation

3.1. Software design goals and overall workflow

Mandala is organized as a layered scientific software stack rather than a single end-to-end training script. The central workflow is: parse electronic-structure outputs into a common sparse representation; harmonize basis conventions; construct graph inputs and matrix targets; train an E(3)-equivariant model; map predictions back to sparse blocks; and evaluate matrix-level and observable-level quantities. This design is reflected in the separation between data abstractions, irrep mapping, network components, observable evaluation, and training orchestration within one coherent codebase.

The practical advantage of this organization is that individual layers can evolve independently. New parser backends do not require changes to the neural network modules. New readout heads do not require a different sparse matrix container. Observable-based loss terms can be added on top of the existing matrix prediction pipeline rather than replacing it. This modularity is one of the principal contributions of the framework and is essential for a project whose scope spans both software engineering and electronic-structure methodology.

3.2. Data model: sparse blocks, snapshots, and basis harmonization

The core data abstractions are `BlockMatrix`, `IrrepsBlockData`, and `Snapshot`. A `BlockMatrix` stores a sparse operator as a dictionary of atom-pair block tensors together with explicit edge metadata indexing lattice shifts and source and destination atoms. A `Snapshot` bundles the matrices belonging to one structure, in particular the Hamiltonian, overlap, and density matrices, along with positions, cell information, and optional force and stress targets. This makes a single object sufficient to support both matrix reconstruction and observable-level evaluation.

The use of a snapshot-level container is crucial for keeping the framework physically coherent. Because H, S, and D correspond to the same structure and basis, they must share atom ordering, orbital metadata, and sparse indexing. Mandala enforces this consistency at construction time and exposes common physics helpers such as the sparse traces used for energy and electron-count evaluation. The same operator-level representation also preserves access to Hamiltonian-based post-processing, including band-structure calculations [TODO: specify workflow/API entry point]. In this way, the matrix prediction problem is formulated at the level of one physically meaningful configuration rather than as three unrelated tensors.

Mandala also includes a dedicated basis-conversion layer. Different electronic-structure packages store localized orbitals in different real spherical-harmonic orderings and sign conventions. The basis converters normalize these conventions to the representation expected by the equivariant learning stack. This harmonization is implemented as a separate software component because it should be reused by parsing, training, and evaluation code alike. This layer supports OpenMX and FHI-aims conventions, and PySCF data are likewise mapped into the same internal sparse format [TODO: finish PySCF training-data integration].

3.3. Parser layer and multi-backend data ingestion

Mandala provides parser support for several electronic-structure data sources. The primary parser target is OpenMX, from which Mandala reads sparse Hamiltonian, overlap, and density blocks together with structural metadata and, when available, forces. Separate parser modules also support FHI-aims outputs and PySCF data [TODO: finish PySCF training-data integration]. Although these backends differ in file format and basis ordering, they are funneled into the same internal `Snapshot` representation.

This unification is a significant practical feature. Electronic-structure machine-learning projects often become tied to a single simulator and a single data layout, which makes comparison, extension, and long-term maintenance difficult. Mandala instead treats file-format handling as a pluggable front end. As long as a backend can provide atomic identities, orbital metadata, positions, and the relevant sparse matrices, it can be incorporated into the same training and evaluation workflow. For a scientific software paper, this separation between backend-specific I/O and backend-agnostic learning logic is especially important because it determines how broadly the code can be reused.

3.4. Irrep mapping and graph construction

The bridge between sparse matrices and equivariant learning is the `BlockIrrepMapper`. For each ordered pair of chemical species, the mapper uses orbital metadata to construct the change-of-basis operators that connect matrix blocks to irreducible E(3) features. This mapping is reused across the data pipeline, the model heads, and the evaluation code, which avoids duplicated representation logic and ensures that training targets and predictions remain aligned.

Graph construction is performed from atomic positions, cell vectors, and species labels. Nodes correspond to atoms. Edges correspond to self-interactions and neighbor interactions within a cutoff radius, with explicit lattice shifts for periodic systems. For each edge, Mandala computes distance embeddings and spherical-harmonic features, and it preserves a canonical ordering that is kept consistent with the sparse matrix representation. The graph and matrix views of the data are therefore coupled but not conflated: the graph is the input geometry on which the neural network operates, while the sparse blocks remain the target operator representation.

3.5. E(3)-equivariant network architecture

The Mandala architecture organizes the model into node encoders, edge encoders, message-passing blocks, and matrix-specific readout heads. Hidden features are represented as direct sums of irreducible representations up to a configurable angular cutoff l_{max}. The edge encoder combines radial information and spherical harmonics, while the message-passing stack updates node and edge features with equivariant tensor-product-based operations. The final readout projects learned edge features back into the irrep spaces associated with the requested matrix blocks and then into sparse matrix form.

Mandala is designed to support a broad architecture search space, not one frozen model. Exposed choices include the hidden irrep content, the number of message-passing layers, tensor-product style, residual pathways, normalization strategy, nonlinearities, edge-update and node-update variants, head depth, the use of tensor-square transformations in the readout, separate treatment of diagonal and shifted-self interactions, and the selection of matrix targets to be predicted jointly. This breadth is one of the major strengths of the framework because it allows systematic experimentation with architecture families while preserving the same data model and training pipeline.

From a software perspective, the multi-head design is especially important. Mandala can be configured to predict only the Hamiltonian or to predict Hamiltonian, overlap, and density simultaneously. Because the heads operate on a shared equivariant latent representation, the cost of multi-target learning is mostly in the final projections rather than in a completely separate model. This structure is what enables operator-level multitask learning and, in turn, the observable-aware training objectives discussed below.

3.6. Observable-aware training on energies, electron counts, and forces

Mandala treats matrix reconstruction as the primary supervised task but augments it with optional losses on observables computed from the predicted matrices. In its most general form, the training objective may be written as

$$\mathcal{L} = \sum_{m \in \mathcal{M}} \lambda_m \mathcal{L}_m^{\text{matrix}} + \lambda_E \mathcal{L}^E + \lambda_N \mathcal{L}^N + \lambda_F \mathcal{L}^F, \tag{5}$$

where $\mathcal{M}$ denotes the set of predicted matrix targets. The matrix terms compare predicted and reference sparse blocks, while the observable terms act on quantities derived from the predicted operators. In the current implementation these include at least

$$E_{\text{pred}} = \mathrm{Tr}(D_{\text{pred}} H_{\text{pred}}), \qquad N_{e,\text{pred}} = \mathrm{Tr}(D_{\text{pred}} S_{\text{pred}}), \tag{6}$$

together with force targets obtained from

$$\boldsymbol{F}_{\text{pred}} = -\frac{\partial E_{\text{pred}}}{\partial \boldsymbol{R}}. \tag{7}$$

This design has several practical benefits. First, observable losses couple the learning of different matrices in a physically meaningful way. The network is not only encouraged to reconstruct each operator individually, but also to do so in a manner that preserves derived observables. Second, the same formulation supports several training regimes: pure matrix regression, multitask matrix-plus-observable regression, and force-aware training derived from the learned electronic representation. Third, because the observables are computed directly from the sparse predicted matrices, no extra surrogate layer is needed between matrix prediction and physical evaluation.

Mandala jointly predicts Hamiltonian, overlap, and density matrices and includes optional losses for energy and electron count. The same framework extends naturally to force supervision obtained through automatic differentiation of predicted energy with respect to positions. These capabilities define the framework's scientific scope and fit naturally into the existing abstractions of the codebase.

3.7. Hyperparameter optimization and architecture search

Architecture and training search are treated as first-class use cases. Mandala uses configuration-driven training together with optional Weights & Biases logging and sweep specifications for rapid experimentation. The search dimensions already explored in the framework include hidden widths, angular cutoffs, radial basis size, message-passing depth, head depth, learning-rate schedules, observable-loss coefficients, force-loss coefficients, symmetry handling, and several alternative equivariant head constructions.

This matters scientifically because sparse matrix learning is not a one-size-fits-all problem. The optimal architecture can depend on the target matrix set, the chemical diversity of the dataset, the desired balance between matrix accuracy and observable accuracy, and the degree to which force supervision should shape the learned representation. By organizing these choices as configurable software parameters rather than ad hoc code edits, Mandala enables systematic model selection and ablation studies without changing the surrounding workflow.

3.8. Verification, reproducibility, and extensibility

Mandala includes a substantial test suite covering data parsing, sparse matrix operations, irrep mapping, graph construction, architectural components, equivariance, force prediction, and end-to-end training workflows. These tests are not incidental; they are part of the framework design. In a code base where basis conventions, sparse indexing, and equivariant tensor algebra interact closely, silent bookkeeping errors are otherwise easy to introduce and hard to diagnose. Explicit unit and physics-style tests therefore play the role of software-level conservation laws.

Reproducibility is supported through configuration-driven training, consistent dataset construction via `DatasetFactory`, explicit checkpointing, and logging hooks that expose both physics metrics and training diagnostics. Extensibility follows from the layered design described above: new file parsers, new equivariant building blocks, new observable definitions, and new training objectives can all be added without discarding the existing infrastructure. This is the software-engineering foundation that allows Mandala to serve both as a research vehicle for new model variants and as a stable framework for future application studies.

4. Conclusion

Mandala is a scientific software framework for learning sparse electronic-structure operators with E(3)-equivariant graph neural networks. Its defining features are the explicit sparse block representation of Hamiltonian, overlap, and density matrices; the modular software layers that connect parsing, basis harmonization, irrep mapping, graph construction, model definition, and training; and the ability to supervise not only matrices themselves but also observables derived from them.

The framework is designed to connect two communities that are often separated in practice. From the electronic-structure side, it retains access to operator-level information and basis-aware quantities. From the atomistic machine-learning side, it inherits the flexibility of modern equivariant message passing, multitask learning, and automated hyperparameter search. By combining these perspectives, Mandala supports workflows in which one can train on matrix fidelity, total-energy fidelity, electron-count fidelity, and force fidelity within one coherent implementation.

Quantitative benchmarks on [TODO: list systems], scientific case studies for [TODO: list applications], and comparative assessments against [TODO: list baseline methods] are reported in Section [TODO: numerical results section]. Mandala is therefore positioned not merely as a prototype model, but as a modular platform for operator-level machine learning in electronic-structure simulation.

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